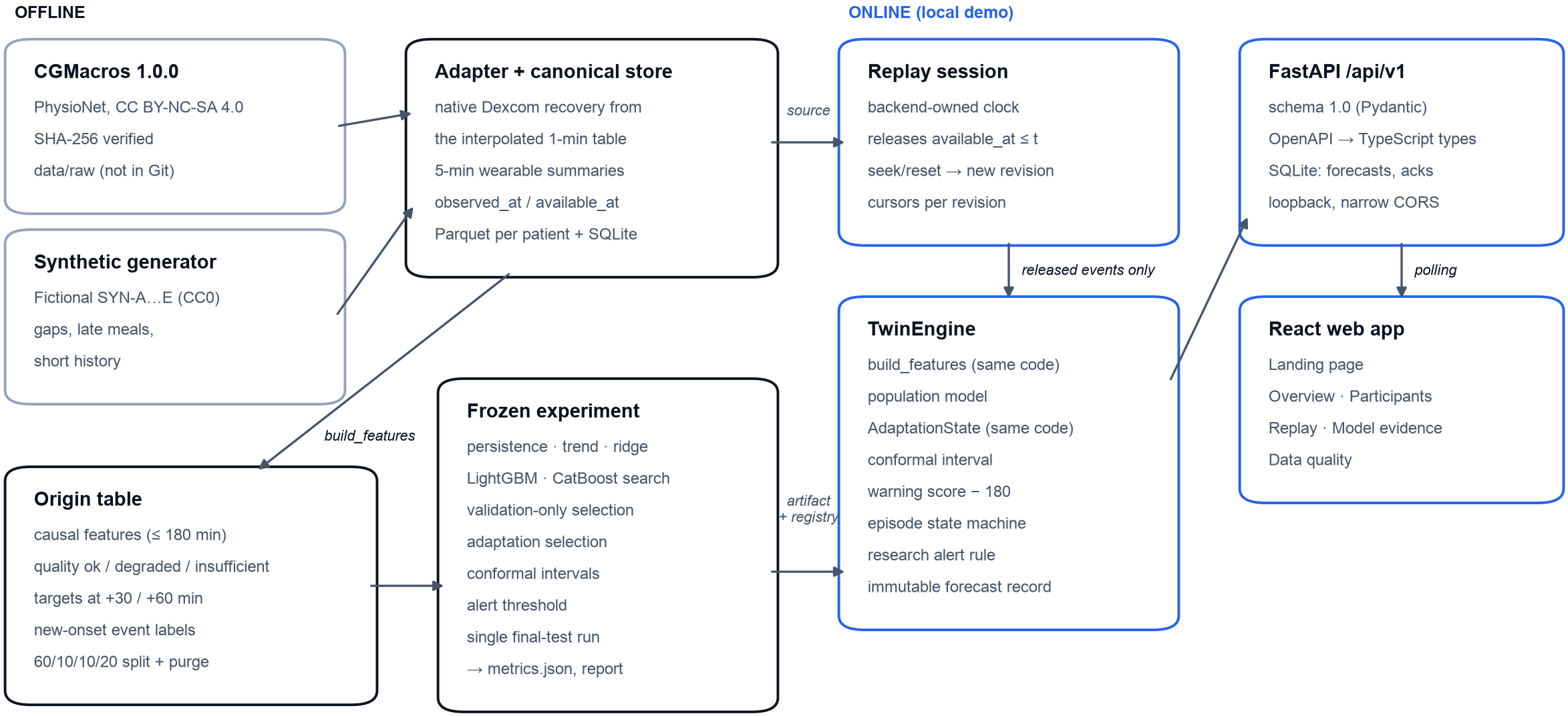
GlucoTwin architecture

Offline: reproducible commands on CPU. Online: local API + browser. The same feature, adaptation and episode code runs in both.



Forecast lifecycle, availability boundary and state ownership

1. Origin. A native glucose reading becomes available at time t. Features use only records with observed_at ≤ t and available_at ≤ t (glucotwin.features.build_features — the same code offline and online).

2. Population model. LightGBM predicts glucose change at +30 and +60 minutes from 38 trailing glucose and wearable features (feature set dyn-v2: each 5-minute wearable summary counts only the part inside the lookback window). Artifacts carry a transformation fingerprint; mismatched weights are refused.

3. Patient adaptation. AdaptationState adds a shrunk, bounded, decaying correction learned only from earlier forecasts whose target readings were already available. Model weights are never refit.

4. Uncertainty and warning. Split-conformal prediction ranges (target coverage 80 %) calibrated for this exact pipeline; warning score = max forecast within 60 min – 180 mg/dL (uncalibrated, not a probability). Warning outcomes use one shared one-to-one onset matcher offline and in replay.

5. Immutable record. Forecasts are stored once under uuid5(session, revision, origin, model version); outcomes are revealed only when the replay clock passes their availability.

6. Revisions. Seek/reset rebuild the twin from released events only; old cursors return 409; the browser keys caches by (session, revision, clock).

State ownership. Canonical store (Parquet + SQLite, local) · replay clock and sessions (API memory) · twin state (per session revision, rebuilt deterministically) · issued forecasts and acknowledgements (SQLite audit) · model artifacts (checksummed registry) · UI preferences (browser).

Posture. CPU only; loopback binding; narrow CORS; no cloud, API keys or LLMs; trusted artifacts verified by SHA-256; no pickles. Research demonstration, not production or a medical device.